

Tutorial Clarity v153 — Sync Fixes Installation Guide

Date: 2026-05-11

Fixes: 3 critical sync bugs causing TTS stutter at ~252 seconds

Base version: v152

Summary of the Problem

In v152, the clarified TTS audio stutters and loses sync with the YouTube video around the 252-second mark. Three root causes were identified:

1. **George can't see the player** — `getVideoTime()` looks for `window.__TC_ACTIVE_YT_PLAYER__` but `page.tsx` stores the player only in a local React ref.
 2. **Synthetic timeline drift** — `adaptSegmentsForAudio()` ignores real YouTube caption timestamps and builds a fake cumulative timeline based on word counts. Any pause, music break, or silence in the video causes progressive drift.
 3. **Always starts at 0:00** — `ClarifyAudioPanel.handleStart()` never passes the current video time to `start()`, so TTS always begins from the beginning regardless of where the user is watching.
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Files Changed (3 files)

FIX #1 — `app/watch/page.tsx`

Install path: `/app/watch/page.tsx`

What changed:

- Added `declare global { interface Window { __TC_ACTIVE_YT_PLAYER__: any; } }` TypeScript declaration at top of file
- In `onReady` callback (YouTube player init), added: `window.__TC_ACTIVE_YT_PLAYER__ = event.target;`
- In cleanup return of the player `useEffect`, added: `window.__TC_ACTIVE_YT_PLAYER__ = null;`

Why: George's `getVideoTime()` in `useAudioTranslation.ts` tries to read `window.__TC_ACTIVE_YT_PLAYER__.getCurrentTime()` to get the real video position. Without this global, George falls back to stale callbacks or estimates, causing drift detection to malfunction.

FIX #2 — `components/useAudioClarification.ts`

Install path: `/components/useAudioClarification.ts`

What changed:

- Rewrote the `adaptSegmentsForAudio()` function (~lines 387-574)
- **REMOVED:** Synthetic cumulative timeline (`let cumulativeTime = 0; ... start = cumulativeTime; cumulativeTime += scaledDuration;`)
- **ADDED:** Resolves original YouTube timestamps (`seg.start`, `seg.duration`, `raw.end`, `raw.startMs/endMs`) and uses them directly
- Falls back to positional estimate only when original timestamps are genuinely missing (NaN/un-

defined)

- Diagnostic logging updated from [v129] to [v153] with timing source tracking

Why: The v129 approach created segment timings like 0.00s, 2.40s, 5.10s... based on word counts when the actual YouTube captions were at 0.00s, 3.50s, 12.80s... (with natural pauses). By 252 seconds, the synthetic timeline had drifted far enough that George's 100ms sync loop entered a stutter cycle of jump→detect-drift→jump.

FIX #3 — components/ClarifyAudioPanel.tsx

Install path: /components/ClarifyAudioPanel.tsx

What changed:

- `handleStart()` now passes `currentTime` to `actions.start(currentTime)` instead of calling `actions.startClarification(videoId, selectedLanguage)` with no time
- Fixed hook destructuring from `{ state, actions }` to `[state, actions]` to match `useClarifyAudio` tuple return
- Fixed method names to match hook API: `stopClarification` → `stop`, `pauseAudio` → `pause`, `resumeAudio` → `resume`, `setVolume` → `setAudioVolume`, `setMuted` → `setAudioMuted`
- Fixed `state.progress` → `state.processingProgress`
- Added default export

Why: The `start()` function in `useClarifyAudio.ts` accepts `startTime?: number` (defaults to 0). Without passing the current video time, TTS always started generating from segment 0, requiring George to fast-forward through potentially minutes of already-watched content.

Installation Steps

1. Back up current files

```
cp app/watch/page.tsx app/watch/page.tsx.v152-backup
cp components/useAudioClarification.ts components/useAudioClarification.ts.v152-backup # if exists
cp components/ClarifyAudioPanel.tsx components/ClarifyAudioPanel.tsx.v152-backup # if exists
```

2. Copy the v153 fixed files

```
# From the v153 package directory:
cp v153/app/watch/page.tsx app/watch/page.tsx
cp v153/components/useAudioClarification.ts components/useAudioClarification.ts
cp v153/components/ClarifyAudioPanel.tsx components/ClarifyAudioPanel.tsx
```

3. Verify the fixes are present

```
# FIX #1: Player exposed globally
grep "__TC_ACTIVE_YT_PLAYER__" app/watch/page.tsx
# Should show 3 lines: declare, assignment in onReady, cleanup in return

# FIX #2: Real timestamps preserved
grep "PRESERVING original YouTube timestamps" components/useAudioClarification.ts
# Should show the v153 log message

# FIX #3: currentTime passed to start
grep "actions.start(currentTime)" components/ClarifyAudioPanel.tsx
# Should show the fixed handleStart call
```

4. Restart the dev server

```
npm run dev
# or
yarn dev
```

5. Test the fixes

1. Open a YouTube tutorial video (ideally 5+ minutes)
2. Seek to ~2:00 in the video
3. Open the Menu → Click “Start Clarification”
4. Verify in browser console:
 - [v153] Player ready – exposed to window.__TC_ACTIVE_YT_PLAYER__ (FIX #1)
 - [v153]  All N segments using original YouTube timestamps (FIX #2)
 - [v153] ClarifyAudioPanel.handleStart() – starting from currentTime=120.00s (FIX #3)
5. Let it play past the 4:12 mark (252 seconds) — stuttering should be eliminated

Support Files (also included)

These files were copied from the v152 uploads into the project structure. They are **unchanged** but were missing from the project directory:

File	Install Path
useClarifyAudio.ts	hooks/useClarifyAudio.ts
useAudioTranslation.ts	hooks/useAudioTranslation.ts
useChunkedTranscription.ts	hooks/useChunkedTranscription.ts
useAudioClarification.ts	hooks/useAudioClarification.ts
useNotifications.ts	hooks/useNotifications.ts
useSpeechRateCalculator.ts	hooks/useSpeechRateCalculator.ts
useVideoProcessing.ts	hooks/useVideoProcessing.ts
useClarificationUsage.ts	hooks/useClarificationUsage.ts
clarifyAudioEngine.ts	lib/clarifyAudioEngine.ts
audioBufferManager.ts	lib/audioBufferManager.ts
voiceAssignment.ts	lib/voiceAssignment.ts
speakerDiarization.ts	lib/speakerDiarization.ts
clientSpeakerDetection.ts	lib/clientSpeakerDetection.ts
cache.ts	lib/cache.ts
ClarificationLimitPopup.tsx	components/ClarificationLimitPopup.tsx

Rollback

To revert to v152:

```
git checkout v152-baseline -- app/watch/page.tsx
git checkout v152-baseline -- components/useAudioClarification.ts
git checkout v152-baseline -- components/ClarifyAudioPanel.tsx
```

Or restore from backups:

```
cp app/watch/page.tsx.v152-backup app/watch/page.tsx
# etc.
```